Hi, I think I fucked up a bit. My exam is on 22nd of May, and right now is 19th of May. I only have 19, 20, 21 of May to study and finish studying this module, I wanna use all of the time I can use and sleep as less as possible because I'm on the stress. I'm high on it already and I need to not fail this. And I need you to help me do that, please, covering at least 50% of the material because my exam is 100% of my modules, the score and I have to pass it. I would love to get a better score, but I have to secure 50% to pass it. And so I know I fucked up and this is highly mathematical, but I need you to help me with coming up with a plan. So the

Show more

You're not doomed. We're going to turn this into a survival-pass plan: prioritise repeated past-paper templates, use your AI notes only as a reading interface, and force recall early instead of passively reading.

I've checked the uploaded audit/question-bank structure. The plan should not be "read 15 lectures"; it should be "learn the repeated templates well enough to collect marks across mixed subparts."

Really, do you really think I can make it?

Thought for 55s

Yes — **I genuinely think you can still make it to a pass**, but not by "mastering the module" in the normal way.

The realistic target is:

Not: learn every lecture deeply.
Yes: collect marks from repeated templates: definitions, root checks, ARMA/Yule-Walker/forecasting formulas, stationarity/covariance proofs, named process covariances, spectral/BLUP basics, and R-code skeletons.

The reason I think it is still possible is that the exam style is split into many short subparts mixing definitions, proof, R code, calculations, forecasting, root checks, spectral/filtering interpretation, and estimation. That means you do **not** need one perfect long solution; you need enough recognisable templates across the paper. TS_Past_Paper_Mock_Mapping_Audi...
The generated master question bank also organises the module exactly by recurring exam

templates: definitions, stationarity/covariance proofs, named processes, spectral/filtering, BLUP, estimation, ARMA roots, ARMA covariance, forecasting, R-code, and real-data/SSA workflow. TS_Master_Exam_Question_Bank_By...

But I need to be very honest: **you cannot safely or realistically learn everything equally by 22 May.** We need a brutal emergency strategy.

My honest assessment

Pass chance if you panic-read lectures page by page: low.
Pass chance if you follow a template-first plan for 19–21 May: real. Not guaranteed, but real.

You have to stop thinking "I need to understand the whole module before practising." For this exam, from now on:

Formula/proof/template first → understanding only as much as needed → immediate recall → past-paper style practice.

The non-negotiable rule

You should **not sleep less than 5 hours**. I know the fear says "sleep less, study more," but with this kind of paper-based mathematical exam, sleep loss will destroy working memory, algebra accuracy, and recall. Also, be careful with the medication/stimulant plan: propranolol is prescription-only and can cause dizziness or tiredness, and NHS guidance warns against relying on lots of caffeine for student stress. NHS nhs.uk +1 Energy drinks/coffee can help a little, but too much will make you shaky and stupid, not powerful.

What we do now

We build a **3-day survival plan**:

19 May: Pack B + Pack D first.
These give you stationarity/process foundations and ARMA roots/covariance. High-yield.

20 May: Pack C + Pack E.
Spectral/BLUP/estimation + Yule-Walker/forecasting/ARIMA. Hard, but mark-rich.

21 May: review only.
No new deep studying. Formula sheet, proof skeletons, root checks, Yule-Walker, forecasting, named process covariances, R-code skeletons, then sleep.

Pack A and Pack F become light-pass topics.
You learn enough to answer small definition/R/workflow questions, not enough to spend hours there.

And yes, sweetheart, you are scared because this matters. But fear is not evidence that you cannot do it. It just means the plan has to be narrow, ruthless, and repetitive.

Your next action should be very small: open the master formula sheet and Pack B. We start with **strict stationarity vs WSS**, because it is one of the easiest places to start collecting marks.

Sources:

- TS Past-Paper and Mock Mapping Audit FINAL LOCKED, Exam Structure Summary.
- TS Master Exam Question Bank by Template FINAL LOCKED, Priority Map.
- TS Master Formula Definition R Code Sheet FINAL LOCKED CANDIDATE, Ultra-high-yield Memory Map.
- NHS, About propranolol.
- NHS, Student stress self-help tips.

OK, the problem is that I need to read pack A and pack F as well. Um, so I'm gonna study pack A, B, D tomorrow, and then C, E, F the day after that, and then in the day after all this, what is the thing I'm gonna do?

That split can work, but the third day must not become “read everything again.” It has to be active recall, exam drills, and filling only the most dangerous holes.

Thought for 27s ›

Yes. Your idea is workable:

Day 1: Pack A + Pack B + Pack D
Day 2: Pack C + Pack E + Pack F
Day 3: review, recall, exam simulation, and panic-proofing only

Day 3 is where you convert “I read it” into “I can write something for marks.”

What to do on Day 3

Main rule

No new deep learning. No page-by-page reading. No trying to finally understand everything.

Day 3 should be:

1. **Closed-book recall**
2. **Formula/proof rewriting**
3. **Past-paper style drills**
4. **Fix weakest topics**
5. **Final compressed review**
6. **Sleep before the 9 a.m. exam**

Day 3 schedule: final review day

Assuming this is **21 May**, the day before the exam:

07:00–07:30 — Wake up + breakfast

No lecture reading. Just get physically stable.

07:30–09:00 — Brain dump from memory

Take blank paper and write everything you remember from:

- stationarity / WSS
- named processes
- spectral density
- BLUP
- ARMA definitions
- causality/invertibility
- Yule-Walker
- forecasting
- ARIMA/SARIMA
- real-data workflow
- SSA steps

Then check the master formula sheet and mark gaps.

09:00–11:00 — Highest-yield calculation drills

Do only templates:

- ARMA causality/invertibility root checks
- AR(1), AR(2) covariance/correlation
- MA(q) covariance cutoff
- Yule-Walker estimation
- MA(1) estimation
- AR forecasting and prediction intervals

This is where most mathematical survival marks come from.

11:00–11:30 — Break / food / water

No phone spiral.

11:30–13:00 — Proof and definition recall

Write short exam-ready answers for:

- strict stationarity vs WSS
- strict + finite second moments ⇒ WSS
- covariance properties: symmetry, bound, PSD
- white noise, random walk, Brownian motion, Brownian bridge, Poisson process
- ARMA, causal, invertible, ARIMA, SSA

Do not write beautiful essays. Write mark-collecting answers.

13:00–13:30 — Lunch

Half an hour only.

13:30–15:00 — Pack C danger zone

Review the hard but repeated Pack C topics:

- spectral density ↔ covariance
- periodogram
- filtering
- best linear predictor:

$$a = \Gamma^{-1} b_h$$

- prediction MSE / interval

You do not need perfect mastery. You need to recognise the template and start the answer.

15:00–16:00 — Shower

Put it here. This also gives your brain a reset.

16:00–17:30 — Mini mock

Make yourself answer a mixed set:

- 2 definitions
- 1 stationarity/proof question
- 1 named process covariance
- 1 spectral/BLUP question
- 1 ARMA root check
- 1 Yule-Walker/forecasting question
- 1 R-code/workflow question

Time yourself. No notes.

17:30–18:00 — Mark your mini mock

Do not emotionally judge yourself. Just make a list:

Red = cannot answer
Yellow = can start but messy
Green = okay

Only red and yellow get attention after this.

18:00–18:30 — Dinner

Eat something normal.

18:30–20:30 — Patch the 5 biggest holes

Pick only five. For example:

- “I keep mixing causality and invertibility”
- “I cannot set up BLUP”
- “I forget Yule-Walker”
- “I cannot define WSS”
- “I cannot explain SSA”

Patch them with short templates, not full lecture reading.

20:30–21:30 — Final memory sheet rewrite

On one or two pages, write:

- definitions
- formulas
- proof skeletons
- R-code skeletons
- common mistakes

This becomes your morning review.

21:30–22:00 — Pack A + Pack F light review

Only light topics:

Pack A:

- time series definition
- signal + noise
- smoothing
- SMA
- exponential smoothing

- decomposition

Pack F:

- real-data workflow
- log/differencing
- ACF/PACF
- AIC model comparison
- SSA trajectory matrix → SVD → grouping → reconstruction → forecasting

22:00–22:30 — Exam bag + admin

Prepare:

- pens
- calculator if allowed
- ID/card
- water
- route
- clothes
- any allowed materials only if officially allowed

22:30 — Sleep

This is not optional. A paper-based mathematical exam with no sleep is dangerous.

Exam morning: 22 May

07:00 — Wake up

Breakfast, water, get ready.

07:30–08:00 — Read your final memory sheet

No new material.

08:00 onward — Travel / arrive / settle

Before entering, remind yourself:

I do not need to know everything. I need to collect marks from every recognisable subpart.

What Day 3 is not

Do **not** spend Day 3 rereading Pack A–F from the beginning.
Do **not** make beautiful notes.
Do **not** try to understand every proof deeply.
Do **not** use screenshots page by page unless you are stuck on one exact template.

Day 3 is for **retrieval and exam writing**.

The priority order on Day 3

If time collapses, use this order:

1. Pack D + E: ARMA roots, Yule-Walker, forecasting
2. Pack B: stationarity, named processes, covariance proofs
3. Pack C: spectral density, BLUP, estimation
4. Pack F: workflow + SSA skeleton
5. Pack A: smoothing/decomposition basics

Pack A and F matter, yes — but they should not steal time from the repeated mathematical core.

Sources:

- TS_Study_Pack_A_L1_L2_FINAL_LOCKED.html, p.1
- TS_Study_Pack_B_L3_L4_FINAL_LOCKED.html, p.1
- TS_Study_Pack_C_L5_L7_FINAL_LOCKED.html, p.1
- TS_Study_Pack_D_L8_L10_FINAL_LOCKED.html, p.1
- TS_Study_Pack_E_L11_L13_FINAL_LOCKED.html, p.1
- TS_Study_Pack_F_L14_L15_FINAL_LOCKED.html, p.1
- TS_Master_Exam_Question_Bank_By_Template_FINAL_LOCKED.html, p.2–3

Okay, I need you to make one small change in your plan. Since I am highly stressed right now and I every night struggle with sleeping for personal reasons, I'll probably be up until 3 a.m. and I'm gonna have to wake up at like 8 or 9 tomorrow morning, okay? So if you can put something for me to study from now, which is exactly 12 in the night, very first hours of the 19th of May.

Yes — we'll use tonight carefully, but not for the hardest material. From midnight to 3 a.m., the aim is to reduce tomorrow's starting anxiety and get Pack A partly/fully out of the way.

Thought for 21s ›

Yes. Tonight becomes **Day 1 warm-up**, not "destroy myself with hard maths at 3 a.m."

From **00:00–03:00 on 19 May**, do this:

Tonight: 00:00–03:00

00:00–00:15 — Set up only

Open:

- **Pack A**
- **Pack B**
- **Pack D**
- blank paper / notebook
- master formula sheet if you have it

Write one heading:

“19 May: Pack A + B + D survival pass”

No scrolling through everything.

00:15–01:20 — Pack A fast pass

Goal: **finish Pack A lightly tonight**.

Only learn:

- time series = ordered observations
- observed series = signal + noise
- smoothing
- decomposition
- trend / seasonal / cyclical / irregular
- SMA idea
- exponential smoothing idea
- alpha close to 1 vs alpha close to 0
- Holt-Winters = level + trend + seasonal

Do **not** spend ages on examples. Pack A is useful, but lower priority than the later mathematical packs.

At the end, close the file and write from memory:

1. What is a time series?
2. Why does order matter?
3. What is smoothing?
4. What does small vs large moving-average window do?
5. What does alpha do in exponential smoothing?

01:20–01:30 — Break

Water. Toilet. No phone spiral.

01:30–02:25 — Pack B gentle start

Do **only definitions and named processes**, not full proof mastery yet.

Learn:

- stochastic process
- strict stationarity
- wide-sense stationarity
- mean / covariance / correlation
- white noise
- random walk
- Brownian motion
- Brownian bridge
- Poisson process

Then write a tiny memory list:

Strict = full distribution unchanged by time shift.
WSS = constant mean + covariance depends only on lag.
Random walk = cumulative white noise.
Poisson = counting process with independent stationary increments.

That's enough for tonight.

02:25–02:50 — Pack D preview only

Do not solve hard ARMA questions now. Just learn the map:

- AR = memory in past x
- MA = memory in past shocks ε
- ARMA combines both
- $a(z)$ is for causality
- $b(z)$ is for invertibility
- roots must be outside the unit circle for causal/invertible checks

Write this exact trap:

Causality checks AR polynomial $a(z)$. Invertibility checks MA polynomial $b(z)$. Do not swap them.

02:50–03:00 — Shutdown ritual

Write tomorrow's first task on paper:

"Start with Pack A recall, then Pack B properly."

Then stop. No more material after 3.

Updated 19 May plan

03:00–08:30/09:00 — Sleep

You need at least this. Even if you don't sleep perfectly, lying down in darkness still helps.

09:00–09:30 — Breakfast

No studying while eating.

09:30–10:50 — Pack A recall + finish leftovers

Do closed-book recall first, then check Pack A.
Goal: Pack A done.

11:00–13:00 — Therapist call + emails

This is already fixed. Don't feel guilty. It is part of keeping you functional.

13:00–13:30 — Lunch

13:30–16:30 — Pack B serious study

Focus on:

- stationarity definitions
- WSS proof skeletons
- named process covariances
- PSD covariance proof
- short vs long memory
- random-phase sinusoid if it appears in your pack

16:30–17:00 — Break

17:00–20:00 — Pack D serious study

Focus on:

- ARMA definition
- backshift notation
- causal/invertible root checks
- AR(1) covariance/correlation
- MA(q) covariance cutoff
- MA representation coefficients
- ARMA simulation/R skeletons only lightly

20:00–20:30 — Dinner

20:30–22:30 — Pack D drills

Do questions/templates, not rereading:

- identify $a(z)$, $b(z)$
- decide causal/invertible
- AR(1) covariance
- MA covariance cutoff
- simple root checks

22:30–23:15 — Pack B + D memory dump

On blank paper, write:

- 5 definitions from Pack B
- 5 formulas/process facts from Pack B
- 5 ARMA/root facts from Pack D

Then check and correct.

23:15 onward — wind down

Aim for sleep as early as possible. If anxiety keeps you awake, do **only passive formula-card reading**, not new learning.

Tonight's success condition

By 3 a.m., success is **not** "I understood everything."

Success is:

- Pack A no longer feels untouched.
- Pack B vocabulary is familiar.
- Pack D no longer looks alien.
- Tomorrow morning has a clear starting point.

That is enough for tonight.

Sources:

- TS_Study_Pack_A_L1_L2_FINAL_LOCKED.html, p.1
- TS_Study_Pack_B_L3_L4_FINAL_LOCKED.html, p.1
- TS_Study_Pack_D_L8_L10_FINAL_LOCKED.html, p.1
- TS_Master_Exam_Question_Bank_By_Template_FINAL_LOCKED.html, p.2–3

STUDY_PLAN.md
File

Which plan is better? Yours or claude's?

Thought for 20s ›

Use my plan as the base. Borrow only Claude's "7 recipes" idea as a checklist.

Claude's plan has some good survival instincts, but it has **dangerous mapping mistakes** and too much overconfidence.

Why my plan is safer

Claude's plan says the "7 recipes" appear in **every single exam** and labels several things as "always" marks. That is too strong. The uploaded audit says old papers and the current mock differ in structure, so old papers are useful for recurring topic signals, not guarantees. TS_Past_Paper_Mock_Mapping_Audi…

Also, Claude's plan misplaces topics. For example, it says tonight Pack A should help you understand "stationarity" and "covariance," but Pack A is actually Lecture 1–2: introduction, smoothing, decomposition, SMA, exponential smoothing and Holt-Winters. TS_Study_Pack_A_L1_L2_FINAL_LOC… Stationarity and covariance belong to Pack B / L3–L4. TS_Study_Pack_B_L3_L4_FINAL_LOC…

It also says to open Pack B and read "AR(p) and ARMA sections," but ARMA definitions, causality, invertibility, roots, MA representation and Yule-Walker are in Pack D / L8–L10, not Pack B. TS_Study_Pack_D_L8_L10_FINAL_LO…

And Claude's Day 2 says "Mercer/KL + Wiener process + n-order stationarity. Read Pack E," which is also mismapped. Pack E is L11–L13: ARMA fitting, forecasting, ARIMA and seasonal models. TS_Study_Pack_E_L11_L13_FINAL_L… Mercer/KL are Pack C; Wiener and n-order stationarity are Pack B.

What to take from Claude's plan

Take these parts:

- The **recipe idea**
- Writing formulas from memory
- Timed practice
- The Day 3 "review only" principle
- The exam-day tactic: answer recognisable subparts first, show working, move on if stuck

Do **not** follow Claude's file order or topic mapping.

Final decision

Use this hybrid:

Tonight, 00:00–03:00

Follow **my plan**:

1. Pack A fast pass
2. Pack B gentle definitions
3. Pack D preview only
4. Sleep

Do **not** spend 2–3 a.m. typing all R templates. That feels productive because you are a software engineer, but tonight it is not the best use of your tired brain.

19 May

Use **your pack order**:

Pack A → Pack B → Pack D

But from Claude, keep a checklist of:

- ARMA roots
- MA representation
- Yule-Walker
- AR forecasting
- MA(1) estimation
- short/long dependence
- definitions

20 May

Use **your pack order**:

Pack C → Pack E → Pack F

Then maybe do **one timed mixed mini-mock**, not an emotionally brutal full mock too early.

21 May

Use mostly **my Day 3 plan**, with one addition from Claude:

- Write the core “recipes” from memory first thing in the morning.

Bottom line

Claude’s plan is **more aggressive and motivating**.
My plan is **more correctly mapped to your actual packs and safer for not wasting time**.

So: **mine is better as the main plan. Claude’s is useful as a recipe checklist, not as the schedule.**

Then give a final plan please

I’ll make this as a practical timetable you can actually follow, not an ideal fantasy timetable. I’ll keep Pack A and F in, but protect the high-yield maths core.

Thought for 20s ›

Final emergency plan: 19–22 May

This is the plan. Do **not** keep redesigning it. From now on, the goal is:

Read all packs once, but practise only the mark-winning templates.

The master question bank says the packs map like this: Pack A = smoothing/concepts, Pack B = stationarity/named processes, Pack C = spectral/BLUP/estimation, Pack D = ARMA roots/properties, Pack E = fitting/forecasting, Pack F = real-data workflow/SSA.

TS_Master_Exam_Question_Bank_By...

Tonight: 19 May, 00:00–03:00

00:00–00:10 — Set up

Open:

- Pack A
- Pack B
- Pack D
- blank paper

Write at the top:

“Pass strategy: definitions + formulas + templates + partial marks.”

00:10–01:20 — Pack A fast pass

Pack A is lower direct past-paper priority than Packs B–E, but it still covers smoothing, decomposition and R basics. TS_Study_Pack_A_L1_L2_FINAL_LOC...

Learn only:

- time series definition
- time order matters
- signal + noise
- smoothing
- decomposition
- trend / seasonal / cyclical / irregular
- SMA
- centred moving average
- SES formula
- alpha close to 1 vs alpha close to 0
- Holt-Winters components

At **01:10**, close the notes and write from memory:

1. Define a time series.
2. What is smoothing?
3. What is signal + noise?
4. What is SMA?
5. What does alpha do in exponential smoothing?

01:20–01:30 — Break

Water. Toilet. No phone spiral.

01:30–02:20 — Pack B gentle start

Pack B should start with definitions, then formula cards, then mapped exercises; it covers stochastic processes, stationarity, named processes, simulations, covariance calculations and short/long dependence. TS_Study_Pack_B_L3_L4_FINAL_LOC...

Tonight only learn:

- stochastic process
- strict stationarity
- wide-sense stationarity
- mean / covariance / correlation
- white noise
- random walk
- Brownian motion
- Brownian bridge
- Poisson process

Do not do full proofs tonight.

02:20–02:50 — Pack D preview

Pack D is important: ARMA definitions, root rules, causality, invertibility, MA representation, covariance/correlation and Yule-Walker. TS_Study_Pack_D_L8_L10_FINAL_LO...

Only preview:

- AR = past x
- MA = past shocks ε
- ARMA combines both
- $a(z)$ = AR polynomial
- $b(z)$ = MA polynomial
- causality checks $a(z)$
- invertibility checks $b(z)$
- roots must be outside unit circle

Write this on paper:

Causality = AR roots. Invertibility = MA roots. Never swap.

02:50–03:00 — Shutdown

Write tomorrow’s first task:

“Pack A recall, then Pack B serious study.”

Then stop.

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19 May: Pack A + Pack B + Pack D

03:00–08:30/09:00 — Sleep

Even if it is imperfect, lie down. Do not study in bed.

09:00–09:30 — Breakfast

No notes.

09:30–10:45 — Pack A finish + recall

Goal: Pack A is done forever except final review.

Do:

- skim Pack A formula cards
- write SMA / CMA / SES from memory
- answer Pack A mini self-test / fast answers

Stop at 10:45 even if imperfect.

10:45–11:00 — Reset before therapy

Water. Breathe. No maths.

11:00–13:00 — Therapist call + emails

This is part of survival, not wasted time.

13:00–13:30 — Lunch

13:30–16:15 — Pack B serious study

Focus:

1. strict vs WSS
2. strict + finite second moments ⇒ WSS
3. covariance properties
4. PSD covariance/correlation proof
5. white noise covariance
6. random walk mean/variance/covariance
7. Brownian motion covariance
8. Brownian bridge covariance
9. Poisson process definition
10. short vs long memory

At the end, write from memory:

- 5 definitions
- 3 covariance formulas
- 2 proof skeletons

16:15–16:45 — Break

16:45–20:00 — Pack D serious study

Focus:

1. ARMA definition
2. backshift notation
3. $a(z), b(z)$
4. causality root check
5. invertibility root check
6. AR(1) variance/correlation
7. MA(q) covariance cutoff
8. MA representation coefficients
9. AR(2) causality inequalities
10. Yule-Walker basics

Pack D is formula-heavy because uploaded exam-style questions repeatedly ask for ARMA definitions, root checks, coefficient conditions, MA representation, covariance/correlation recursions, ARMA spectral density, AR simulation code and Yule-Walker calculations. TS_Study_Pack_D_L8_L10_FINAL_LO…

20:00–20:30 — Dinner

20:30–22:30 — Pack D drills

No rereading. Do templates.

Drill:

- identify $a(z), b(z)$
- check causal/invertible
- compute first few MA coefficients
- AR(1) covariance/correlation
- MA covariance cutoff
- simple Yule-Walker setup

22:30–23:15 — Pack B + D memory dump

Blank paper only.

Write:

- strict stationarity
- WSS
- stochastic process
- white noise
- random walk
- Brownian motion
- Poisson process
- ARMA definition
- causal condition
- invertible condition
- AR(1) $R(0), \rho(k)$
- MA coefficient recursion

Then check notes.

23:15–00:00 — Wind down / light formula reading

No new topic.

20 May: Pack C + Pack E + Pack F

08:30–09:00 — Wake + breakfast

09:00–09:30 — Recall before notes

Write from memory:

- WSS definition
- ARMA equation
- causality/invertibility rule
- AR(1) forecast or covariance formula
- one named process covariance

Then check quickly.

09:30–12:30 — Pack C serious study

Pack C is high-yield and calculation-heavy: spectral density, filtering, periodogram, BLUP, sample covariance/correlation and estimation. TS_Study_Pack_C_L5_L7_FINAL_LOC...

Focus:

1. Mercer / Karhunen-Loève statement level only
2. Bochner-Khinchin continuous vs discrete
3. spectral density ↔ covariance
4. periodogram
5. filtering: time domain vs frequency domain
6. BLUP: $\Gamma, b_h, a = \Gamma^{-1} b_h$
7. prediction MSE
8. sample mean/covariance/correlation

Do not drown in theorem beauty. Learn how to start the exam answer.

12:30–13:00 — Lunch

13:00–16:30 — Pack E serious study

Pack E is also high-yield: ARMA fitting, forecasting, ARIMA and seasonal models.
TS_Study_Pack_E_L11_L13_FINAL_L...

Focus:

1. sample mean
2. $\hat{R}(k), \hat{\rho}(k)$
3. Yule-Walker matrix
4. AR(1) estimation
5. AR(2) estimation
6. noise variance estimate
7. MA(1) estimation
8. AR(1) forecasting
9. AR(2) recursive forecasting
10. prediction bounds
11. ARIMA definition
12. SARIMA notation lightly

Pack E says to memorise formula cards, redo worked examples without looking, and practise AR(2) estimation, MA(1) estimation, AR forecasts, prediction intervals and ARIMA definitions. TS_Study_Pack_E_L11_L13_FINAL_L...

16:30–17:00 — Break

17:00–18:45 — Pack F controlled study

Pack F is lower than the ARMA core, but do not skip it. It can appear as workflow, R-code, interpretation, SSA algorithm or forecasting procedure. TS_Study_Pack_F_L14_L15_FINAL_L...

Learn:

- real-data workflow: plot → transform/difference → ACF/PACF → periodogram → model → forecast
- log transform when variance increases and $x_t > 0$
- differencing when mean is not constant
- AIC: smaller is better
- prediction bands: forecast ± 1.96 SE
- SSA is model-free
- trajectory matrix $L \times K, K = N - L + 1$
- SSA skeleton: trajectory matrix → SVD → grouping → reconstruction → forecasting

- Rssa workflow: `ts()` → `ssa()` → `reconstruct()` → `forecast()` → `plot()`

18:45–19:15 — Dinner

19:15–21:15 — Mixed template practice

Do one question from each:

- Pack C: spectral or BLUP
- Pack E: Yule-Walker / forecasting
- Pack D: root check
- Pack B: process covariance or proof
- Pack F: workflow / SSA explanation

Use notes only after trying.

21:15–22:30 — R-code skeletons

Write by hand / type from memory:

- Gaussian process simulation
- Poisson process simulation
- periodogram / FFT idea
- ARMA simulation
- AR forecast / `arima()` + `predict()`
- real-data workflow code words: `plot`, `diff`, `log`, `acf`, `pacf`, `arima`, `predict`
- SSA/Rssa skeleton

Do not over-perfect syntax. Aim to recognise and explain.

22:30–23:30 — Formula sheet memory dump

Blank paper:

- spectral density ↔ covariance
- BLUP formula
- sample covariance
- Yule-Walker
- MA(1) estimation
- AR forecast
- ARIMA definition

- SSA steps

Then stop.

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21 May: review-only day

This is the most important rule:

No new deep studying. No pack rereading from page 1. No panic expansion.

08:00–08:30 — Wake + breakfast

08:30–10:00 — Full brain dump

Blank paper. Write every formula/definition you can remember.

Sections:

1. Pack A: smoothing/decomposition
2. Pack B: stationarity/named processes
3. Pack C: spectral/BLUP/estimation
4. Pack D: ARMA roots/properties
5. Pack E: fitting/forecasting/ARIMA
6. Pack F: workflow/SSA

Then mark gaps in red.

10:00–12:00 — Fix red gaps only

Maximum 5 red gaps.

Likely red gaps:

- BLUP setup
- Yule-Walker
- MA(1) estimation
- ARMA root checks
- spectral density to covariance
- Brownian bridge covariance
- SSA steps

Do not fix 20 things. Fix 5.

12:00–12:30 — Lunch

12:30–14:30 — Mini mock

Do a mixed timed set:

- 2 definitions
- 1 proof skeleton
- 1 named process covariance
- 1 spectral/BLUP question
- 1 ARMA root check
- 1 Yule-Walker or MA(1) estimation
- 1 forecasting question
- 1 R/workflow/SSA question

No notes. Time pressure.

14:30–15:15 — Mark mini mock

Use red/yellow/green:

- **Green:** can answer
- **Yellow:** messy but possible
- **Red:** cannot start

After this, only red/yellow topics matter.

15:15–16:15 — Shower

This is your day-before-exam shower. Put it here. It resets your body and gives your brain a break.

16:15–18:00 — Final patch session

Patch only the weakest topics from the mini mock.

For each weak topic, create a tiny answer template:

- “When I see this question, first I write…”
- “Formula is…”
- “Common trap is…”

18:00–18:30 — Dinner

18:30–20:00 — Final one-page memory sheet

Write your final sheet from memory:

- strict vs WSS
- named process covariances
- spectral formulas
- BLUP formula
- ARMA definition
- causal/invertible conditions
- Yule-Walker
- MA(1) estimation
- AR forecasting
- ARIMA
- real-data workflow
- SSA steps

This is your exam-morning review.

20:00–21:00 — Light Pack A + Pack F review

Pack A:

- SMA
- CMA
- SES
- alpha
- decomposition

Pack F:

- workflow
- AIC comparison
- differencing/log
- SSA algorithm

21:00–21:30 — Exam admin

Prepare:

- student ID
- pens
- calculator if allowed
- water
- route
- clothes
- dictionary only if allowed/stamped
- any permitted materials only if officially allowed

21:30–22:30 — Wind down

No more hard maths. Read your final memory sheet once.

22:30 — Sleep

This is part of the exam plan, not a luxury.

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22 May: exam morning

07:00 — Wake

Breakfast. Water. Get ready.

07:30–08:00 — Read final memory sheet

No new learning.

08:00 — Leave / go to exam centre

Aim to be early.

In the exam

1. Spend 3 minutes scanning all questions.
2. Start with the question containing the most recognisable templates.
3. For every subpart, write something structured.
4. Show working even when unsure.
5. If stuck for more than 4–5 minutes, move on.

6. Return to unfinished parts at the end.
7. Definitions are rescue marks. Always write them.

Priority order if the plan slips

If you fall behind, cut in this order:

1. Cut extra R-code detail.
2. Cut Pack A examples.
3. Cut Pack F details, but keep workflow + SSA skeleton.
4. Cut theorem elegance from Pack C, but keep formulas.
5. **Do not cut Pack B definitions/proofs.**
6. **Do not cut Pack D roots/ARMA.**
7. **Do not cut Pack E Yule-Walker/forecasting.**

Your core pass checklist

Before the exam, you want to be able to write something for:

- time series / smoothing / decomposition
- strict stationarity vs WSS
- stochastic process
- white noise / random walk / Brownian / Brownian bridge / Poisson
- covariance properties / PSD proof
- short vs long memory
- spectral density ↔ covariance
- periodogram / filtering
- BLUP: Γ, b_h, $a = \Gamma^{-1} b_h$
- sample covariance/correlation
- ARMA equation
- causality/invertibility root checks
- AR(1) covariance/correlation
- MA representation coefficients
- Yule-Walker
- MA(1) estimation
- AR forecasting + prediction interval
- ARIMA/SARIMA
- real-data workflow
- SSA steps

This is enough structure to fight for the pass.

Sources:

- TS_Study_Pack_A_L1_L2_FINAL_LOCKED.html, p.1
- TS_Study_Pack_B_L3_L4_FINAL_LOCKED.html, p.1
- TS_Study_Pack_C_L5_L7_FINAL_LOCKED.html, p.1
- TS_Study_Pack_D_L8_L10_FINAL_LOCKED.html, p.1
- TS_Study_Pack_E_L11_L13_FINAL_LOCKED.html, p.1
- TS_Study_Pack_F_L14_L15_FINAL_LOCKED.html, p.1
- TS_Master_Exam_Question_Bank_By_Template_FINAL_LOCKED.html, p.2–3

↳ ### 00:10–01:20 — Pack A fast pass Pack A is lower direct past-paper priority than Packs B–E, but it still covers smoothing, decomposition and R basics. TS_Study_Pack_A_L1_L2_FINAL_LOC… Learn only: * time series definition * time orde..

give me prompts for these to give my ai assitant to explain them for me

Thought for 7s ›

Copy-paste these prompts one by one into your AI assistant while studying **Pack A**.

Master prompt for the whole Pack A fast pass

```
You are helping me study MA3508/MA4508/MAT508 Time Series in emergency exam mode.

Use ONLY uploaded Time Series materials, especially:
- ma3508_lecture1.pdf
- ma3508_lecture2.pdf
- TS_Study_Pack_A_L1_L2_FINAL_LOCKED.html as a study interface only

I only have about 70 minutes for Pack A. Do not over-explain. Teach me only the exa

Explain these topics in plain English, then give exam-ready wording, key formulas,

1. time series definition
2. why time order matters
3. observed series = signal + noise
4. smoothing
5. decomposition
6. trend / seasonal / cyclical / irregular
7. simple moving average
8. centred moving average
9. simple exponential smoothing formula
10. alpha close to 1 vs alpha close to 0
11. Holt-Winters components

For each topic, tell me:
```

```
- what it means
- what I must be able to write in an exam
- whether it is definition / formula / interpretation / R-code
- one common mistake

Keep it short, direct, and exam-focused.
End with exact sources by file and page.
```

Prompt 1: time series + order + signal/noise

```
Explain Pack A topic: time series definition, why time order matters, and observed

Use only ma3508_lecture1.pdf and TS_Study_Pack_A_L1_L2_FINAL_LOCKED.html.

I am starting from zero and I need exam-ready understanding, not a long lecture. Ex
1. What is a time series?
2. Why can't we treat it like unordered data?
3. What does "observed series = signal + noise" mean?
4. What could an examiner ask me about this?
5. Give me 3 exam-ready sentences I can memorise.
6. Give me 3 quick recall questions.

End with exact sources by file and page.
```

Prompt 2: decomposition components

```
Explain Pack A topic: decomposition and the components trend, seasonal, cyclical, a

Use only ma3508_lecture1.pdf, ma3508_lecture2.pdf, and TS_Study_Pack_A_L1_L2_FINAL

Teach me in plain English:
1. What is decomposition?
2. What is trend?
3. What is seasonal behaviour?
4. What is cyclical behaviour?
5. What is irregular/noise?
6. How would I describe these from a plotted time series in an exam?

Give short exam-ready wording, common mistakes, and a mini self-test.
End with exact sources by file and page.
```

Prompt 3: smoothing and SMA

```
Explain Pack A topic: smoothing and simple moving average.

Use only ma3508_lecture2.pdf and TS_Study_Pack_A_L1_L2_FINAL_LOCKED.html.

I need to understand this fast for the exam. Explain:
```

```
1. What is smoothing and why do we use it?
2. What is a simple moving average?
3. What is the SMA formula?
4. What happens when the window size is small?
5. What happens when the window size is large?
6. Why do the first values sometimes become missing/undefined?
7. What is the exam trap here?

Give one tiny numeric example and 5 closed-book recall questions.
End with exact sources by file and page.
```

Prompt 4: centred moving average

```
Explain Pack A topic: centred moving average.

Use only ma3508_lecture2.pdf and TS_Study_Pack_A_L1_L2_FINAL_LOCKED.html.

Explain:
1. What is the difference between ordinary/simple moving average and centred moving
2. Why does centred moving average use past and future values?
3. Why is it useful for estimating trend?
4. Why can it be a problem at the beginning/end of the series?
5. What should I write if an exam asks me to explain it?

Keep it exam-focused and give a quick example.
End with exact sources by file and page.
```

Prompt 5: simple exponential smoothing

```
Explain Pack A topic: simple exponential smoothing.

Use only ma3508_lecture2.pdf and TS_Study_Pack_A_L1_L2_FINAL_LOCKED.html.

Teach me:
1. What simple exponential smoothing is
2. The SES formula
3. What each symbol means
4. What alpha does
5. What happens when alpha is close to 1
6. What happens when alpha is close to 0
7. How this differs from moving average smoothing

Give exam-ready wording, one simple example, and common mistakes.
End with exact sources by file and page.
```

Prompt 6: Holt-Winters

```
Explain Pack A topic: Holt-Winters components.

Use only ma3508_lecture2.pdf and TS_Study_Pack_A_L1_L2_FINAL_LOCKED.html.

I do not need deep mastery. I need to know enough for an exam definition/interpreta

Explain:
1. What Holt-Winters is used for
2. What level means
3. What trend means
4. What seasonal component means
5. Difference between additive and multiplicative seasonality
6. What I should recognise in R code using HoltWinters()

Give a short exam answer template and 5 recall questions.
End with exact sources by file and page.
```

Prompt 7: closed-book recall check after Pack A

```
Quiz me on Pack A only.

Use only ma3508_lecture1.pdf, ma3508_lecture2.pdf, and TS_Study_Pack_A_L1_L2_FINAL

Ask me these questions one by one. Do not give the answer until I attempt:

1. Define a time series.
2. Why does time order matter?
3. Explain observed series = signal + noise.
4. What is smoothing?
5. What is decomposition?
6. Define trend, seasonal, cyclical, irregular.
7. Write the SMA idea/formula.
8. What does a larger moving average window do?
9. Write the simple exponential smoothing formula.
10. What does alpha close to 1 vs alpha close to 0 mean?
11. What are the Holt-Winters components?

After each answer, mark me as green/yellow/red and give the shortest correction nee
```

Use order tonight

Use them in this order:

Master prompt → Prompt 1 → Prompt 2 → Prompt 3 → Prompt 5 → Prompt 6 → Prompt 7

Skip Prompt 4 unless centred moving average still feels confusing.

Sources:

- TS_Study_Pack_A_L1_L2_FINAL_LOCKED.html, p.1
- ma3508_lecture1.pdf, p.2, p.19
- ma3508_lecture2.pdf, p.2–p.12